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Machine Learning-Driven Groundwater Potential Prediction: A Data Driven Approach

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***Abstract*— Groundwater is a vital resource supporting global water supply, agriculture, and industry. Accurately predicting groundwater potential is essential for effective management, yet traditional hydrological models often struggle with the complexity of groundwater dynamics. This study leverages machine learning (ML) techniques to enhance groundwater prediction, comparing Support Vector Regression (SVR), Random Forest (RF), and Logistic Regression (LR) models. Among these, RF demonstrated superior performance with an accuracy of 84.5%. A feature importance analysis identified precipitation as the most influential factor, followed by slope and elevation. Furthermore, groundwater trends for 2026–2035 were forecasted, revealing significant fluctuations and a sharp decline around 2032–2033. Model validation using confusion matrices and regression analysis confirmed the reliability of predictions. The findings provide a data-driven approach for policymakers and hydrologists to optimize groundwater management strategies, ensuring sustainability in the face of climate change and increasing demand.**

Keywords*— *groundwater, artificial intelligence , machine learning , groundwater potential, AI modelling

I. Introduction

Groundwater is a vital natural resource, supplying approximately 50% of the world's drinking water and sustaining 40% of global irrigation systems [1]. It plays a crucial role across agricultural, industrial, and domestic sectors, making the accurate assessment and prediction of groundwater availability essential for sustainable water management. While traditional hydrological models have served as the foundation for groundwater studies, they often struggle to capture the complex, nonlinear interactions governing groundwater dynamics [2]. Recent advancements in machine learning (ML) have revolutionized groundwater modelling by offering robust predictive capabilities and adaptability to diverse hydrogeological conditions [3]. ML-based approaches provide enhanced accuracy in groundwater potential mapping, recharge estimation, and depletion forecasting, allowing researchers and policymakers to develop data-driven water management strategies [4]. Furthermore, ML models can integrate spatial and temporal datasets, enabling the identification of critical groundwater influencing factors and optimizing conservation efforts[5]. Pham et al. introduced GIS-based ML models, such as Support Vector Machines (SVM), for groundwater potential mapping. Their findings emphasized the significance of topographic and hydrological factors, including slope, elevation, terrain ruggedness index (TRI), and height above the nearest drainage (HAND), in predicting groundwater distribution [6]. The application of ensemble ML models has also demonstrated significant improvements in groundwater assessments. A study conducted in West Bengal, India, employed Random Forest (RF), Adaptive Boosting (AdaBoost), Extreme Gradient Boosting (XGBoost), and Voting Ensemble (VE) models to delineate groundwater potential zones. The results revealed that RF and XGBoost outperformed other techniques, highlighting their reliability in regions with diverse hydrogeological conditions [7]. Additionally, hybrid ML models, which integrate deep learning and optimization algorithms, have shown promising results in groundwater level forecasting. Studies incorporating Deep Neural Networks (DNN) with optimization techniques, such as the Artificial Cultures Optimization Algorithm (AVOA), have demonstrated high predictive accuracy, with AUC values exceeding 0.95 in some instances [8]. These advancements underscore the growing role of ML in groundwater research, providing new perspectives on sustainable groundwater resource management.

Contribution of the Study

- **Model Comparison** – Assessed multiple ML models (SVR, RF, LR) for groundwater prediction, with Random Forest emerging as the best-performing model (84.5% accuracy).
- **Feature Importance Analysis** – Identified key factors like precipitation, slope, and elevation, with precipitation being the most influential.
- **Future Groundwater Prediction (2026-2035)** – Forecasted groundwater trends, highlighting fluctuations and a sharp decline around 2032-2033.
- **Performance Evaluation** – Used confusion matrices and regression analysis to validate model accuracy, ensuring reliable predictions.
- **Practical Impact** – Provides a data-driven approach for policymakers and hydrologists to enhance groundwater management and sustainability.

II. Review of Literature

The application of machine learning in groundwater studies has been widely explored in recent years. A review assessed over 200 studies utilizing ML techniques for groundwater quality modelling, emphasizing ML's ability to predict pollution risks associated with chemical contaminants [9]. The review highlighted ML's capability in analysing large datasets, uncovering complex patterns, and facilitating groundwater management decisions.

A study evaluated Artificial Neural Networks (ANN), Adaptive Neuro-Fuzzy Inference Systems (ANFIS), Support Vector Regression (SVR), and Deep Learning models. They emphasized key methodological considerations such as dataset size, variable selection, time-step intervals, and performance evaluation metrics, which are essential for

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optimizing ML-based groundwater predictions [10]. Several ML models have demonstrated strong predictive performance in groundwater assessments. Patidar et al. (2021) integrated geospatial and ML techniques, including Classification and Regression Trees (CART), Boosted Regression Trees (BRT), and Random Forest (RF), to create groundwater potential maps for the Shipra River Basin, India. Their results confirmed that ML-based models effectively handle spatial data complexities, making them suitable for groundwater potential mapping [11].

Ensemble learning approaches have also been widely adopted for groundwater resource analysis. A study in drought-prone regions of Eastern India applied bagging and boosting ensemble ML techniques, including RF, AdaBoost, XGBoost, and VE, to predict groundwater availability. Findings demonstrated that ensemble models improved accuracy and addressed spatial data challenges [12].

Deep learning models, such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, have also been utilized for temporal and spatial groundwater predictions. These models effectively capture nonlinear dependencies and hydrogeological patterns, improving forecasting accuracy [13]. However, challenges such as interpretability, high computational costs, and data scarcity remain critical concerns in ML-based groundwater modelling [14].

Despite these advancements, several challenges persist. ML models require large, high-quality datasets, which may not always be available, particularly in developing regions [15]. Additionally, black-box ML models pose interpretability issues, limiting their practical adoption by water resource managers and policymakers [16]. Future research must enhance model transparency, integrate ML with physically based hydrological models, and develop cost-effective methods for data acquisition and processing [17].

III. Methodology

The input data sets as represented in the Figure 1, serve as the primary data for the pre-processing. These pre-processing methods also include the feature selection process. The data are then processed for training then to categorize them as Training and Extraction. The trained models as and output is the Support Vector Regression (SVR), Random Forest (AF) and Linear Regression (LR).

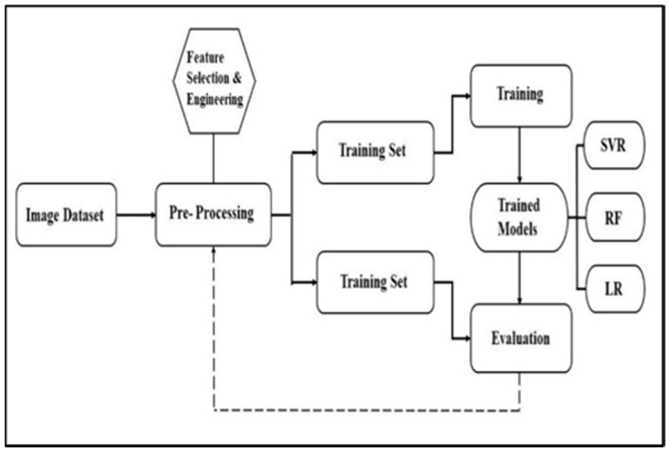


Figure 1: Overall methodology for the proposed framework

A. Datasets

This study utilizes a synthetic groundwater potential dataset designed to assess groundwater availability based on various hydrogeological and environmental parameters [18]. The dataset comprises 10 independent variables, including Elevation, Slope, Soil Type, Precipitation, Drainage, Geology, Land Use, Vegetation, Lithology, and Aquifer characteristics, which are key factors influencing groundwater occurrence. The dependent variable, Groundwater Potential, represents the target measure, indicating the likelihood of groundwater presence in a given region.

The dataset was generated using a combination of simulated hydrogeological models and data augmentation techniques. The base data was derived from real-world studies on groundwater potential, incorporating observed geological and environmental properties. To ensure statistical robustness and diversity, synthetic data augmentation techniques such as Gaussian noise addition, feature interpolation, and domain-specific sampling were applied. This approach enhances the dataset's variability and allows for improved generalization in predictive modelling tasks.

Table 1: Various environmental and hydrological dataset

Elevation (m)	93.27	26.45	14.99	8.99	56.34
Slope (°)	61.06	11.93	92.4	7.77	93
Soil Type	95.88	86.23	16.79	32.68	17.25
Precipitation (mm)	89.83	30.44	50.01	42.24	17.18
Drainage (mm/hr)	23.38	7.99	0.06	60.83	36.21
Geology	68.2	33.75	38.15	73.47	92.78
Land Use	31.92	42.42	49.24	85.64	52.32
Vegetation	8.97	64.27	15.54	41.44	62.91
Lithology	65.55	76.77	49.45	49.45	4.38
Aquifer	46.42	32.81	40.47	82.26	59.35
Groundwater Potential	86.08	35.7	30.96	24.35	56.01

B. ML Models and Training

To evaluate the predictive capabilities of different machine learning models for groundwater potential assessment, three algorithms—Support Vector Regression (SVR), Random Forest (RF), and Logistic Regression (LR)—were trained and tested using hydrogeological datasets. The dataset included critical spatial and meteorological parameters such as precipitation, slope, elevation, land use, soil type, and groundwater depth, which were normalized using Min-Max scaling to ensure uniform feature distribution. Feature selection was performed using the Recursive Feature Elimination (RFE) method to identify the most influential

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variables. The SVR model was trained using the Radial Basis Function (RBF) kernel with a regularization parameter (C) of 1.0 and an epsilon value of 0.1. The Random Forest model was configured with 100 decision trees, a maximum depth of 10, and a minimum sample split of 2, utilizing the Gini index for feature selection. Logistic Regression was trained using the LBFGS solver with an L2 penalty and a regularization parameter of 1.0. A 5-fold cross-validation approach was implemented to ensure model robustness, and performance evaluation was conducted using precision, recall, and F1-score, with Random Forest achieving the highest values across these metrics. This demonstrates its effectiveness in handling high-dimensional data and capturing complex nonlinear relationships in groundwater potential prediction.

C. Feature Selection

For finding the feature selection and ground water potential prediction we are using each model to compare the features like Aquifer, Lithology, Vegetation, Land use, Drainage, Precipitation, Soil type, Slope and Elevation with the importance. We can find from the below graphical representation (Figure 2) that the value comparing to the others precipitation is leading.

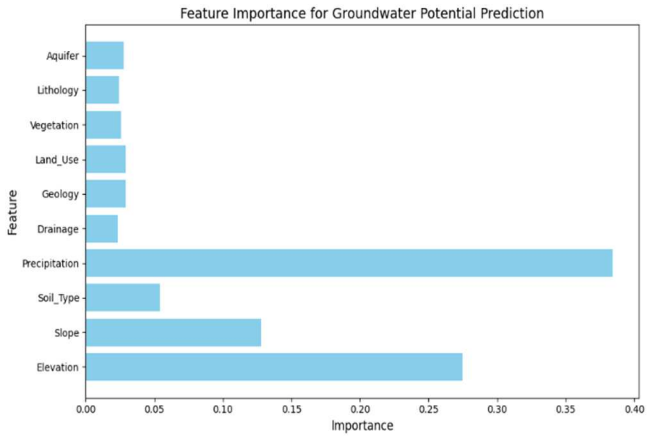


Figure 2: Feature Importance for Groundwater Potential Prediction

Further optimisation is made by comparing the relative feature values such as Precipitation, Slope and Elevation and presented in Figure 3.

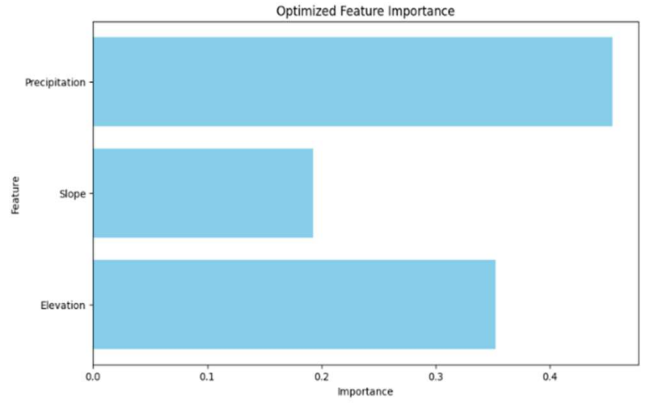


Figure 3: Optimized Feature Importance

Further optimisation also shows high value for the importance of precipitation in each model.

D. Implementation

The LR regression model shows and accuracy of 0.85 for the values comparing with the other models. Below is the data that was produced to the model for running the Logistic regression analysis as shown in Figure 4.

```
Logistic Regression
Accuracy: 0.85
Confusion Matrix:
 [[84 16]
 [13 87]]
Classification Report:
              precision    recall  f1-score   support

           0       0.87      0.84      0.85       100
           1       0.84      0.87      0.86       100

    accuracy                           0.85       200
   macro avg       0.86      0.85      0.85       200
weighted avg       0.86      0.85      0.85       200
```

Figure 4: LR Regression Model showing accuracy of 0.85

The SVM regression model shows and accuracy of 0.84 for the values comparing with the other models. Below is the data that was produced to the model for running the Support vector classifier analysis as shown in Figure 5.

```
Support Vector Classifier
Accuracy: 0.84
Confusion Matrix:
 [[82 18]
 [13 87]]
Classification Report:
              precision    recall  f1-score   support

           0       0.86      0.82      0.84       100
           1       0.83      0.87      0.85       100

    accuracy                           0.84       200
   macro avg       0.85      0.84      0.84       200
weighted avg       0.85      0.84      0.84       200
```

Figure 5: SVC Regression Model showing accuracy of 0.84

The RF regression model shows and accuracy of 0.85 for the values comparing with the other models. Below is the data that was produced to the model for running the Rain Forest Classifier analysis as shown in Figure 6

```
Random Forest Classifier
Accuracy: 0.85
Confusion Matrix:
 [[83 17]
 [13 87]]
Classification Report:
              precision    recall  f1-score   support

           0       0.86      0.83      0.85       100
           1       0.84      0.87      0.85       100

    accuracy                           0.85       200
   macro avg       0.85      0.85      0.85       200
weighted avg       0.85      0.85      0.85       200
```

Figure 6: RF Regression Model showing accuracy of 0.85

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IV. Results And Discussion

Figure 7 analysis implies the insights about the Random Forest Classifier's performance. It suggests that the model perform well and interprets as:

- True Negatives = 83
- False Positives = 17
- False Negatives = 13
- True Positives = 87

The colour intensity indicates that the dark blue suggests higher number of predictions and lighter blue suggests errors

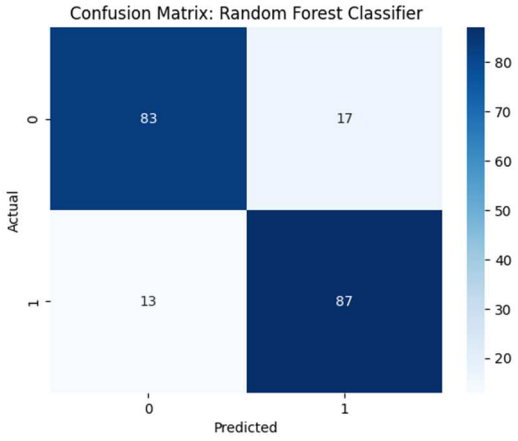


Figure 7: Confusion matrix values.

The Optimized Random Forest Classifier has a balanced precision with 84.5% accuracy. The accuracy measures overall correctness as shown in Figure 8.

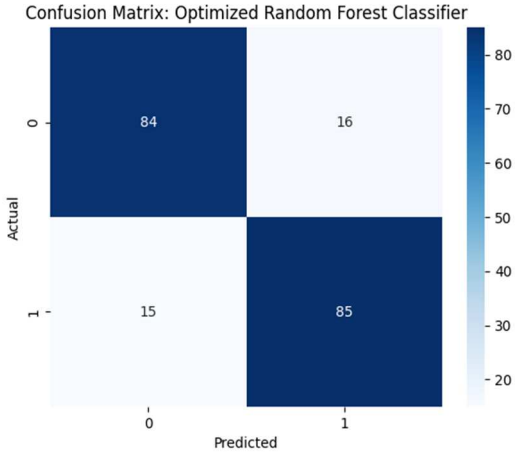


Figure 8: Performance of an Optimized Random Forest Classifier

Figure 9 , this is a scatter plot which represents Regression Predictions vs Actuals for groundwater potential. The X-axis represents the Actual Groundwater potential and Y-axis represents the Predicted Groundwater Potential. The blue dots show the difference between the actual groundwater potential and the expected values. A perfect prediction line is shown by the red line. The model is accurate if the blue dots are near the red line. If the blue dots deviate, which means the model has errors.

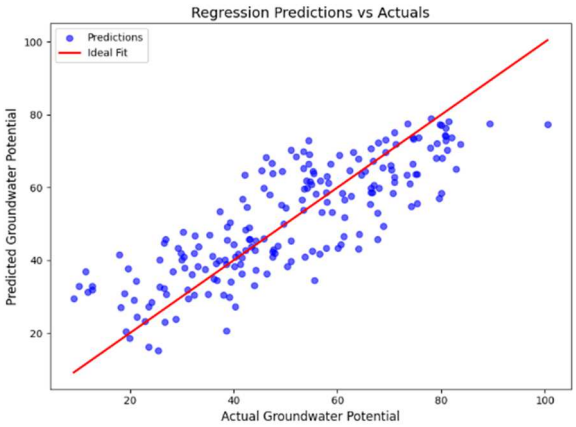


Figure 9: Regression Predictions vs Actuals

In this graph, the X-axis represents the Year and Y- axis represents the Predicted Groundwater Potential. It overall gives a fluctuating trend. The predicted groundwater potential shows significant variations over time. It peaks in 2026 (0.8), then falls in 2027, rises once more by 2029-2030 (0.85) and then sharply declines around 2031.Around 2032-2033 (0.1-0.2) seems to be the lowest point, before it recovers slightly in 2034-2035 (0.5). as depicted in Figure 10

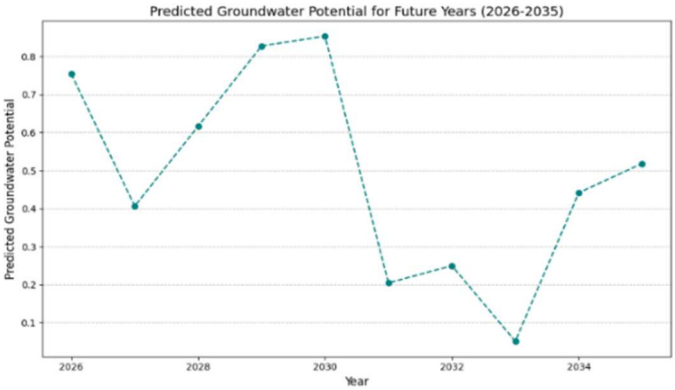


Figure 10: Predicted Groundwater Potential for Future Years (2026-2035)

V. Conclusion

The Random Forest Regressor was chosen for its robustness in handling mixed data types and its ability to rank feature importance. For classification tasks, Logistic Regression, Support Vector Classifier, and Random Forest Classifier were analysed. These models provided insights into binary classification based on groundwater potential. The Random Forest model excelled in both regression and classification tasks. The study provides major contribution for the future references and research is, Model Comparison – Assessed multiple ML models (SVR, RF, LR) for groundwater prediction, with Random Forest emerging as the best-performing model (84.5% accuracy). Feature Importance Analysis – Identified key factors like precipitation, slope, and elevation, with precipitation being the most influential. Future Groundwater Prediction (2026-2035) – Forecasted groundwater trends, highlighting fluctuations and a sharp decline around 2032-2033. Performance Evaluation – Used confusion matrices and regression analysis to validate model accuracy, ensuring reliable predictions. Practical Impact – Provides a data-driven

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approach for policymakers and hydrologists to enhance groundwater management and sustainability.

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